

GROUP 6

SERVICE MONITORING SYSTEM WITH MONIT





GROUP 6



GROUP MEMBER

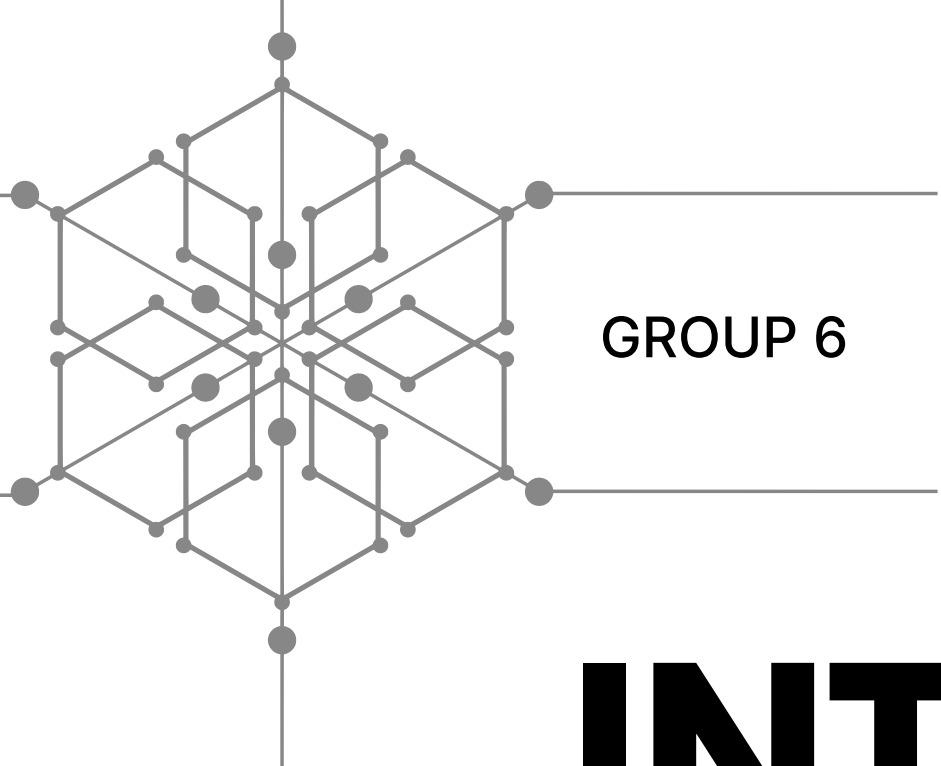
Nguyễn Phạm Trí	CE192008
Bùi Gia Kính	DE181001
Dương Đình Quốc Bảo	SE140034



TABLE OF CONTENT

- 1. Introduction**
- 2. Monit – Overview**
- 3. Installation & Basic Setup**
- 4. Advanced System & Service
Monitoring with Monit**
- 5. Centralized Monitoring with
M/Monit**
- 6. Conclusion**





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INTRODUCTION

- Problem: service failure → system disruption
- Solution: need automatic monitoring tool
- Monit: lightweight, versatile, fast response

MONIT – OVERVIEW

- OPEN-SOURCE, LIGHTWEIGHT
- PERIODIC CHECKS (EVERY CYCLE)
- FLEXIBLE CONFIGURATION
- WEB GUI + EMAIL ALERT SUPPORT

Monit Service Manager

Monit is running on freebsd80-x64 with *uptime, 4d 14h 28m* and monitoring:

	Status	Load			CPU	Memory
4	Running	[0.06]	[0.02]	[0.00]	0.1%us, 0.1%sy	14.1% [34.3 MB]

	Status	Uptime	CPU Total	Mem
	Running	4d 14h 27m	0.0%	1
	Running	40d 21h 42m	0.0%	3
	Running	40d 21h 42m	0.0%	2
	Running	40d 21h 42m	0.0%	0
	Running	40d 21h 42m	0.0%	0.1
	Running	40d 21h 42m	0.0%	0
	Running	40d 21h 42m	0.0%	0
	Running	40d 21h 42m	0.0%	24

INSTALLATION & BASIC SETUP

- **Install via YUM or source**
- **Configure email alerts**
- **Web interface (port 2812)**
- **Allow specific IPs, enable SSL**
- **Add services to /etc/monit.d/**



ADVANCED SYSTEM & SERVICE MONITORING WITH MONIT

1. PROCESS MONITORING

CHECK PROCESS <NAME> PIDFILE <PATH> HOẶC MATCHING <REGEX>

EXAMPLE - NGINX:

```
check process nginx with pidfile /run/nginx.pid
start "/usr/bin/systemctl start nginx.service"
stop  "/usr/bin/systemctl stop nginx.service"
if failed port 80 protocol http then restart
alert <your-email>@gmail.com
```

EXAMPLE - SCRIPT (NO PID):

```
CHECK PROCESS bash_script MATCHING "bash.sh"
START = "/bin/bash /usr/local/bin/bash.sh &"
STOP  = "/usr/bin/pkill -f bash.sh"
IF DOES NOT EXIST THEN RESTART
IF 3 RESTARTS WITHIN 5 CYCLES THEN TIMEOUT
```

2. FILE / DIRECTORY / FILESYSTEM MONITORING

FILE:

```
CHECK PROGRAM ssh_config_checker WITH PATH  
"/usr/local/bin/check_sshd.sh"  
IF STATUS != 0 THEN ALERT
```

FILESYSTEM:

```
CHECK FILESYSTEM rootfs WITH PATH /  
IF SPACE > 80% THEN ALERT  
IF INODE > 90% THEN ALERT
```

DIRECTORY:

```
check directory nginx_log_dir with path /var/log/nginx  
if failed permission 755 then alert  
if failed uid 0 or gid 0 then alert
```


3. PROGRAM/SCRIPT MONITORING

- **EXAMPLE - DISK USAGE SCRIPT:**

check program check_disk_space with path
"/usr/local/bin/check_disk.sh"if status != 0 then alert

```
check program check_disk_space with path "/usr/local/bin/check_disk.sh"  
if status != 0 then alert
```

```
#  
█
```

```
^G Help
```

```
^X Exit
```

```
^O Write Out
```

```
^R Read File
```

```
^W Where Is
```

```
^_ Replace
```

```
^K Cut
```

```
^U Paste
```

```
^T Execute
```

```
^J Justify
```

```
^C Location
```

```
^_ Go To Line
```

4. SYSTEM & SERVICE MONITORING

- **SYSTEM RESOURCE:**

check system localhost

if loadavg (1min) > 2 then alert

if memory usage > 80% then alert

if cpu usage (user) > 70% then alert

- **NETWORK INTERFACE:**

check network net0 with interface ens33

if failed link or changed link then alert

- **SERVICE (SENDMAIL, APACHE):**

check process apache with pidfile /var/run/httpd/httpd.pid

start = "/bin/systemctl start httpd"

stop = "/bin/systemctl stop httpd"if cpu > 60% for 2 cycles then restart

if totalmem > 200 MB for 2 cycles then restart

if 3 restarts within 5 cycles then timeout

CENTRALIZED MONITORING WITH M/MONIT

- **M/MONIT SERVER SETUP:**

```
WGET HTTPS://MMONIT.COM/DIST/MMONIT-4.3.4-LINUX-X64.TAR.GZ  
TAR -XZF MMONIT-4.3.4-LINUX-X64.TAR.GZ  
LN -S MONIT-4.3.4-LINUX-X64 MMONIT
```

- **OPEN PORT 8080:**

```
FIREWALL-CMD --PERMANENT --ADD-PORT=8080/TCP  
FIREWALL-CMD --RELOAD
```

- **MONIT AGENT CONFIGURATION:**

```
SET MMONIT HTTP://ADMIN:SWORDFISH@<M-MONIT-IP>:8080/COLLECTOR  
MONIT -T && MONIT RELOAD
```

- **ACCESS THE INTERFACE:**

```
HTTP://<M-MONIT-IP>:8080
```

CONCLUSION

- **Early issue detection, fast recovery**
- **Reduce downtime**
- **Lightweight, easy, flexible**
- **Less manual work for admins**

VRQUEST

THANK YOU!

GROUP 6